Support and Rehabilitation of Patients with Pulmonary Expansion Deficit by Using Game Therapy

P. F. S. Chacon, C. F. Schon, V. H. L. A. Furtado, G. L. A. M. Signoretti, J. P. P. Oliveira, A. G. Ribeiro, C. D. V. Wanderley, A. A. R. Diniz and H. B. Soares.

Abstract— **Patients suffering from hypoventilation and pulmonary expansion deficit are at increased risk of developing pulmonary complications such as atelectasis, pneumonia or pleural effusion. These complications can increase the length of stay and spending on health, and generate long-term functional impairment. This study aims to produce a therapeutic alternative to the traditional method of lung re-expansion through incentive spirometry (IS) using the game therapy to build an innovative system. This system makes use of infrared and Bluetooth communication technology to associate the game therapy to EI. At the end of the system implementation, we expect to obtain good adhesion of the patient and the physiotherapists.**

I. Introduction

Lung expansion or reexpansion therapies (LET) were created from the need of treating or preventing the reduction of lung volume. Lung collapse happens as cause or consequence of volumetric loss in lungs and can take to hypoxemia or increase the risk of infections or lung injury, if this condition could not be reverted [1]. In postoperative of cardiac or upper abdominal surgeries, it was reported a rate of pulmonary complications between 17 to 88%. Among people with neuromuscular diseases, there is a report of great loss in respiratory muscles responsible for the pressure gradient needed for lung expansion [2].

The application of expansive maneuvers can reduce by 50% the risk of the pulmonary complications mentioned before, what emphasizes the clinical importance of this therapy [7].

These LETs change the transpulmonary pressure gradient, which is obtained by the difference between the alveolar and pleural pressures. The higher this gradient, the greater is the pulmonary expansion and, hence, greater the inspiratory volume. The inspiratory capacity (IC) is the maximum volume of air that can be inspired after a baseline expiration and tidal volume (VT) is the volume of air which enters and leaves the lungs of a normal breath. This it is what is expected to be improved by applying this therapy [3, 13, 14].

There are many ways of applying this therapeutic approach to minimize consequences of a deficit in lung expansion. The incentive spirometry (IS) is one of the wide applied techniques, which provides a visual feedback to encourage patients to perform maximum sustained inspiration, promoting greated lung insufflation [6]. By this way, there is a growing alveolar recruiting, contributing to maintain the alveolar permeability, improving its ventilation, thus, the gas exchange. This is the mechanism that justifies the application of IE for the restauration of lung volume, and preventing respiratory complications [8]. IE is especially beneficial after some surgeries and neuromuscular disorders, such as hemiparesis –, besides being effective after myocardial revascularization [4].

The conventional therapy of incentive spirometry counts on two more usual models, which are defined according to their activation standard: volumetric incentive spirometer (VIS) and flow incentive spirometer (FIS). This second one counts on a nozzle and a hose as interface between the patient and the equipment. During its use, it is necessary that the patient perform maximum inspiration in a way that the spheres within the equipment slightly and uniformly ascend. Besides that, inspirations have to be alternated with some short intervals of rest, which are commonly 5 to 15 repetitions, per session, in each hour, while the patient is awake [11].

Besides being routinely used in clinical practices, its potentiality can be reduced by the lack of cooperation and acceptance of the therapy. These problems can be verified because of the difficulty in coordinating ventilation and possible pain caused by respiratory muscles, which can be compromised, resulting in a bad use of the equipment [5].

The application of games can be a viable option to change this situation, as they are giving good results when applied to the physiotherapy of neurologic and locomotor system disorders [9]. The current context of games development includes not only the entertainment function, but also their use to solve problems, where their application in health stands out among the most promising ones, and being defined as 'serious games'. Those serious games are used in health, because they improve the motivation, favours distracting the patient during painful therapies, and make easier performing therapeutic exercises (exergames), enabling the physiotherapist to rehabilitate the patient in practical clinic (game therapy) [10].

The development of this game therapy by using the flow spirometer becomes more viable, once that this equipment can be sterilized and presents a considerably low cost when compared to the volumetric spirometer.

From the good evidences found in literature about the application of game therapy, and the need of developing some stimulating method to support the treatment of the patients attended by the Hospital Universitário Onofre Lopes (HUOL), at the Federal University of Rio Grande do Norte (UFRN), it was developed a system named ExpGame,

*Research supported by LABSIM/UFRN.

All the authors are with the Federal University of Rio Grande do Norte, Caixa Postal 1524, Campus Universitario Lagoa Nova, Natal/RN, CEP: 59.078-970, Brazil (corresponding author to provide phone: +55-84-99986-4066; e-mail: helianabs@gmail.com).

combining the game and the lung expansion therapy, aiming at improving the efficiency of this kind of treatment, and adding statistic modules to evaluate the data captured during the therapy.

This paper is organized as follows: section II describes materials and methods applied for developing this research; section III presents its description and implementation; section IV illustrates the obtained results by using the software, and points out future works.

II. Materials and Methods

A. The Proposed Model

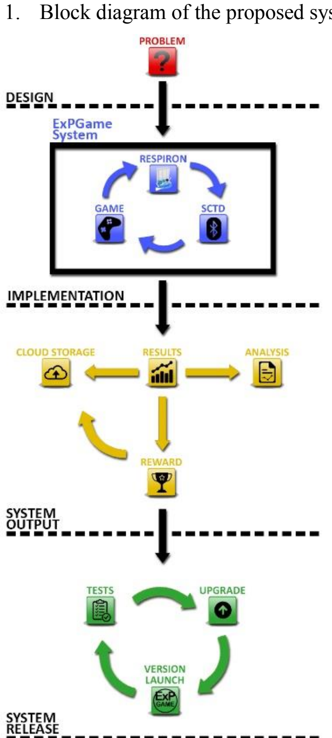
To support the therapeutic treatment of lung expansion in children, young people, adults, and elderly people, it was combined the game therapy with the incentive spirometry. By this way, it was developed a game named ExpGame, which is composed by a system of data capture and transmission, connected to a software.

To accomplish that goal, we designed and implemented a two parts system named ExpGame. The system is composed by a serious game, that works as a user interface between the patient and the spirometer, and a device to capture and transmit the data from spirometer.

The game was thought to show a score to motivate the player as well as useful data for the physiotherapist to analyze. These results could then lead to a reward for the patient and also be stored in a database or cloud storage.

The next step corresponds to launch a preliminary version for testing and, based on the feedback, make the necessary upgrades and roll out a new version. This cycle repeats until we get to a system that fully meets all the criteria needed. That proposed model is presented in the diagram block showed in Fig. 1, where is possible seeing all the developed steps.

Figure 1. Block diagram of the proposed system.



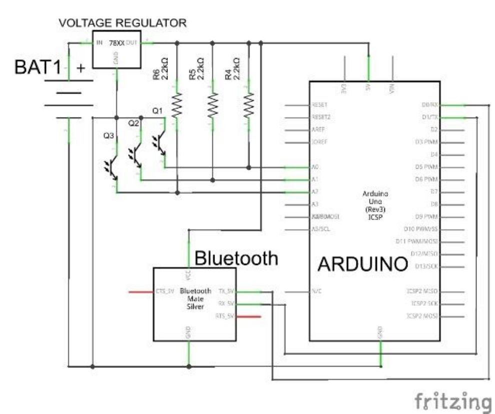
B. The ExpGame

The process of system development began by analyzing the requirements to comprehend the objectives of the equipment and therapy dynamics. From this analysis, it was defined all the basic criteria for the proper use of game therapy as proposed by Maggiorini [10]. It was also considered that each patient could have specific needs, including the possibility of configurating the game variables (repetitions and level of difficulty, among others); giving feedback about the patient performance; ways of motivating the patient during the game; and reports presenting information about each patient, allowing physiotherapist to perform clinical analysis [9].

Air rate, time and volume were selected as essential variables for the proposed system. From this resolution, it was investigated proper tools to measure them. Besides that, it was observed the need of inputting some specific data (weight and name) and presenting some expected results (inspiratory time and estimated inspiratory capacity). Therefore, the game was developed and also the capturing and transmitting data system (CTDS).

The CTDS is responsible for capturing and transmitting all the data about inspiratory time of each patient and uses infrared sensors coupled to the superior part of the incentive spirometer (IS). The infrared sensor is of active type, composed by an infrared light emitting diode and a phototransistor that detects and reacts for that kind of light. That sensor detects the presence of the spheres within the IS, by the interruption of the IR light when they move. It happens because, when the user reaches some inspiratory flow levels, those spheres reach the highest part of the equipment and interrupt the light from the emissor to the receiver. When that light stops coming, the phototransistor changes its output voltage that was being received by the Atmega328p, changing its input level. By detecting this change, that microcontroller process this information and send it to the game through a Bluetooth module (Fig. 2).

Figure 2. Circuit schematic for the proposed system.



The game was programmed in C#, through the Unit 5 platform, combined with the Microsoft Visual Studio Community 2013 IDE [15]. It was designed to have three difficulty levels:

- Easy – patients have to keep the first sphere on the top of the equipment, for 2 seconds;

- Medium –patients have to keep two spheres on the top, for 3 seconds;
- Hard – patients have to keep all the spheres on the top, for 3 seconds.

If the patient is successful, independent from the difficulty level, the stone will be loaded to the catapult, launched and hit the castle. If it does not happen, the stone will be launched with a little strength, and will not hit the castle.

After each successful try, points will be added to the patient score and the castle will seem more damaged. By this way, it is expected that the patient feels stimulated to give his best performance in pulmonary expansion.

The game is divided in 3 main screens (or scenes, as they are referred in the Unity editor): Title, Options, Game, as follows:

- Title: it is the screen where the game is initialized (Fig.3). It presents the game logo and fields to make possible introducing the name and age of the patient;
- Options: this is the screen for choosing the game`s options (Fig. 4), where the physiotherapist can configure the system according to patient's needs. It can be configured: level of difficulty (easy, medium, hard); number of repetitions (5, 10, or 15); and number of series (1, 2, or 3);
- Game: it is the main screen of the game (Fig. 5), where the catapult is located at the left side and the castle at the right side. The upper and lower parts of the screen present data about the game (player, level of difficulty, remaining repetitions, number of series and score). Also in the upper part, there is a power bar, which changes according to the difficulty level, and is fulfilled when the patient keeps the sphere at the top of the equipment. The stone is launched when this bar is completely fulfilled.

After the initial screen (Fig. 3), the game shows a configuration screen (Fig. 4), which allows the physiotherapist to configure the system according to his needs. It can be configured: level of difficulty (easy, medium, hard); number of repetitions (5, 10, or 15); and number os series (1, 2, or 3).

Figure 3. The title screen.

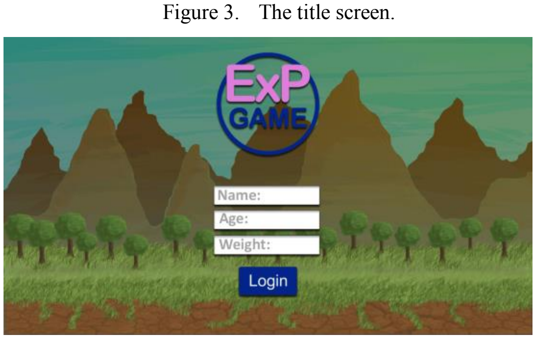


Figure 4. The options screen.

After finishing all the configuration, the game begins (Fig. 5). During the game, all the configured data for that patient is showed on the screen. Moreover, a power bar presents the sustained inspiratory time and it is also showed the score reached by the patient cumulatively after launching each stone.

Figure 5. The game screen.

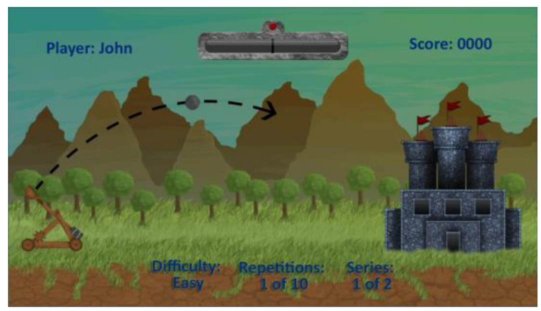
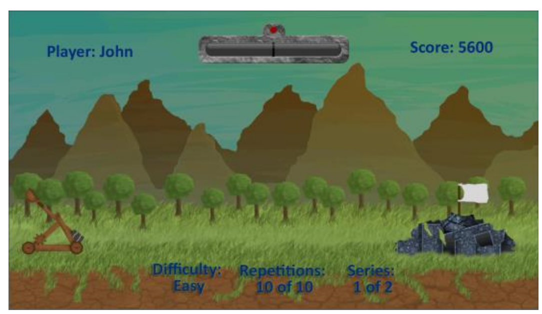


Fig. 6 is showed after finishing a series, presented the castle completely destroyed.

Figure 6. The game screen after a series of launches.



After concluding all the series, the software provides a results screen, showing the patient's cumulated score, during performed series, and a final report to the physiotherapist. On that screen, it is possible observing two inspiratory capacities: measured inspiratory capacity (MIC) and predicted inspiratory capacity (PIC).

The MIC is obtained by knowing that rising up the sphere of the first compartment demands a minimum air rate (Q) of 600 ml/min; for the second compartment, it demands 900 m/min; and for the third compartment, 1,200 ml/min [12], and that the CTDS can measure the interruption time (t) of each chamber, given by:

$$MIC=Q*t \qquad (1)$$

On the other side, the PIC is obtained by summing predicted current volume (PCV) with the inspiratory reserve volume (IRV), as shown in Eq. 3:

$$IRV=6*PCV \tag{2}$$

$$PIC=IRV+PCV \tag{3}$$

Moreover, the PCV is calculated in the following way [13]:

$$PCV=7\ ([ml])/([kg])*weight\ [kg] \tag{4}$$

After finishing these two analysis, their values are graphically shown on the results screen, allowing the physiotherapist to compare them. Those values of inspiratory capacities are stored in the cloud, to allow the physiotherapist to access all that data in future.

III. Conclusion

Game therapy has given encouraging results, especially because of the possibility of using the game to stimulate the patient, what is more difficult in traditional physioterapy, because patients perform repetitive and monotonous movements with weights and equipment.

Another key factor is the low cost of the equipment. It was stipulated that the cost of development of the first prototype was around $60.00 (sixty dollars). This price includes all the components and the cost of the time that was required to build it. As we move forward, the price of the finished equipment can be even smaller if we can mass produce it. Therefore, it is an extremely affordable therapeutic equipment for clinics, hospitals and even the patients themselves.

Indicated for children, young people, adults, and elderly people, the therapy by using videogames does not replace other therapeutic practices, but complements the treatment, once that it takes the patient to other reality, making exercises more fun, and presenting evidences that its results are faster than traditional approaches. The current implementation of the system is aimed for adults but it can be adjusted to fit other age groups. To do that, the pulmonary volume parameters would need to be normalized for each age group.

There are countless recommendations for the application of games to rehabilitate patients; therefore, game therapy can be used to incentive the brain activity and help the patients recover lost moviments. It can be seen not only as a game or therapy technique, but as an improvement that can be combined to therapy techniques.

The developed ExpGame system meets all the requirements defined previously define in its design, being able to promote the interaction between the IS and the game created to the patients performing that kind of therapy. Besides its application as treatment tool, this system was also able to measure the time and, hence, the inspiratory capacity of the patient during the game. This calculation can provide data to evaluate the level of lung expansion obtained by the patient.

This study is currently in its final development stage but it was not yet tested on real users. That way, by this time, there are no practical results to validate the system. The next step of the study is the clinical application of the system in patients who need this kind of therapy.

Acknowledgment

Authors thank to the Laboratory of Signals and Medical Images (LabSIM) and the Federal University of Rio Grande do Norte (UFRN).

References

[1] E. E. T.D. França, F. Ferrari, P. Fernandes, R. Cavalcanti, A. Duarte, B. P. Martinez, E.E. Aquim and M. C. P. Damasceno, "Fisioterapia em pacientes críticos adultos: recomendações do Departamento de Fisioterapia da Associação de Medicina Intensiva Brasileira", Rev. bras. ter. intensiva, vol. 24, no. 1, pp. 6-22, 2012.

[2] T. J. Overend, C. M. Anderson, S. D. Lucy, C. Bhatia, B. I. Jonsson and C. Timmermans, "The Effect of Incentive Spirometry on Postoperative Pulmonary Complications", Chest, vol. 120, no. 3, pp. 971-978, 2001.

[3] V. G. Balista, "PhysioJoy: Sistema de Realidade Virtual para Avaliação e Reabilitação de Déficit Motor", in Joint Conference ICEC 2013, Pollyana Notargiacomo Mustaro, 2013, pp. 16-20.

[4] G. M. Ferreira , M. P. Haeffner , S. M. Barreto and P. Dall'Ago, "Incentive spirometry with expiratory positive airway pressure brings benefits after myocardial revascularization", Arq. Bras. Cardiol., no. 942, pp. 246-251, 2016.

[5] L. L. Hsu, B. K. Batts and J. L. Rau, "Positive Expiratory Pressure Device Acceptance by Hospitalized Children With Sickle Cell Disease Is Comparable to Incentive Spirometry", Respir Care, vol. 50, no. 5, pp. 624-627, 2005.

[6] C. A.C. Azeredo, Fisioterapia respiratoria no hospital geral. Sao Paulo: Manole, 2000.

[7] C. Perrin, J. N. Unterborn, C. D. Ambrosio and N. S. Hill, "Pulmonary complications of chronic neuromuscular diseases and their management", Muscle & Nerve, vol. 29, no. 1, pp. 5-27, 2003.

[8] G. M. Tomich, D. C. França, A.C. M. Diório, R. R. Britto, R. F. Sampaio and V. F. Parreira, "Breathing pattern, thoracoabdominal motion and muscular activity during three breathing exercises", Braz J Med Biol Res, vol. 40, no. 10, pp. 1409-1417, 2007.

[9] C. I. B. Brito, L. V. Brandelli, V. B. Azambuja, "Os efeitos de uma sessão de gameterapia na coordenação motora de indivíduos saudáveis," 2014.

[10] D. Deponti, D. Maggiorini and C. E. Palazzi, "Smartphone's physiatric serious game," Serious Games and Applications for Health (SeGAH), 2011 IEEE 1st International Conference on, Braga, 2011, pp. 1-8.

[11] R. D. Restrepo, K. R. Hirst, L. Wittnebel and R. Wettstein, "AARC Clinical Practice Guideline: Transcutaneous Monitoring of Carbon Dioxide and Oxygen: 2012", Respiratory Care, vol. 57, no. 11, pp. 1955-1962, 2012..

[12] L. Williams and Wilkins, Illustrated manual of nursing practice. Philadelphia:, 2002.,pp. 153-155.

[13] L. Arcêncio, M. D. Souza , B. S. Bortolin, A. C. Fernandes, A. J. Rodrigues and P. R. Evora, "Pre-and postoperative care in cardiothoracic surgery: a physiotherapeutic approach," Rev Bras Cir Cardiovasc, vol. 23, no. 3, pp. 400-410, 2008.

[14] A. C. Guyton and J. E. Hall, Tratado de fisiología médica. Madrid: Elsevier España, 2006, ch. 37, pp. 469.

[15] S. Blackman, Beginning 3D Game Development with Unity 4. Berkeley: California, 2013.

[16] J. Williams, "Narrow-band analyzer (Thesis or Dissertation style)," Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, 1993.

[17] N. Kawasaki, "Parametric study of thermal and chemical nonequilibrium nozzle flow," M.S. thesis, Dept. Electron. Eng., Osaka Univ., Osaka, Japan, 1993.

J. P. Wilkinson, "Nonlinear resonant circuit devices (Patent style)," U.S. Patent 3 624 12, July 16, 1990.